

PROBLEM SET 3

COM SCI 188: Intro to Robotics

Spring 2025

Question:	1	2	3	4	5	Total
Points:	17	17	22	22	22	100

1. Conv Nets.

(a) (5 points) Connect each term with the description of its main purpose.

- | | |
|--------------------|---|
| Conv. Layer • | • Increase non-linearity of the network |
| Pooling Layer • | • Downsample feature maps for efficiency |
| FC Layer • | • Extract features from the input |
| Activation Layer • | • Map extracted features to final predictions |
| Loss Function • | • Evaluate error for classification or regression |

(b) (4 points) What is the main purpose of the "forward pass" in a CNN?

- A) Update the weights
- B) Calculate the output predictions
- C) Measure the loss
- D) Perform gradient descent

(c) (4 points) What happens when the Conv layer uses a stride greater than 1?

- A) It increases image size
- B) It downsamples the image
- C) It makes the network deeper
- D) It adds more weights

(d) (4 points) What is often true about the spatial information in FC layers?

- A) Spatial information is preserved
- B) Spatial information is lost
- C) Spatial information is enhanced
- D) FC layers do not affect spatial information

2. **Particle Filter vs Kalman Filter.** Both Particle Filters and Kalman Filters are built around Bayesian filtering:

- Prediction: predict where you might be after motion.
- Update: correct the prediction using sensor measurements.
- Belief representation: (particles for PF, mean/covariance for KF).

(a) (10 points) For each concept below, choose whether it applies to Particle filter, Kalman filter, or both.

Concept	PF	KF
Represents belief with multiple samples (particles)	☐	☐
Assumes Gaussian noise and linear system	☐	☐
Uses motion models and measurement models	☐	☐
Handles non-linear and non-Gaussian models naturally	☐	☐
Uses a single mean and covariance matrix	☐	☐
Performs prediction-update (Bayes filtering cycle)	☐	☐
May require resampling to focus on high-probability particles	☐	☐
Uses an optimal gain to balance prediction and measurement	☐	☐
Requires an initial estimate (initial belief)	☐	☐
Can be used for robot localization and SLAM	☐	☐

(b) (4 points) In robot localization, what does the measurement model represent when applying Bayes' Theorem?

- A) The probability of the robot moving to a new location given a control input.
- B) The probability of receiving a sensor measurement given the robot's current location.
- C) The probability of the robot's initial position.
- D) The probability of a landmark being detected at any point on the map.

(c) (3 points) Why is probabilistic reasoning important for robotics?
(provide a 1-2 sentence short answer)

3. Simultaneous Localization and Mapping.

- (a) (4 points) What is the primary purpose of SLAM in robotics?
- A) To control robot movements and avoid obstacles
 - B) To detect and track moving objects in the environment
 - C) To map an unknown environment while tracking the robot's position
 - D) To optimize battery usage during navigation
- (b) (4 points) Why can't a basic Kalman Filter (KF) be used directly for SLAM?
- A) SLAM systems have no noise.
 - B) SLAM motion and measurement models are nonlinear.
 - C) Landmark motions are not Gaussian.
 - D) Kalman Filters require extensive compute.
- (c) (4 points) Which of the following is true about the Extended Kalman Filter (EKF)?
- A) It assumes the system is perfectly linear.
 - B) It uses Jacobians to linearize nonlinear models.
 - C) It cannot be used for robot localization.
 - D) It eliminates all sensor noise.
- (d) (10 points) A robot moves according to the nonlinear motion model:

$$g(x, y, \theta, d) = \begin{bmatrix} x + d \cos(\theta) \\ y + d \sin(\theta) \\ \theta \end{bmatrix}$$

where d is the forward distance traveled. Suppose at a given time:
 $x = 2.0$, $y = 3.0$, $\theta = 30^\circ$ (in radians, $\theta = \pi/6$), and $d = 1.0$ meter.

1. Write the Jacobian G_t of the motion model with respect to (x, y, θ) .

2. Compute the numerical values of the non-zero elements of G_t at this state.
(Use $\sin(\pi/6) = 0.5$ and $\cos(\pi/6) = 0.866$ for calculations.)

4. Motion Planning.

- (a) (2 points) What is the primary objective of motion planning?
 - A) To find a sequence of control inputs that maximizes the robot's speed.
 - B) To compute a collision-free path from a start to a goal position.
 - C) To minimize the robot's battery consumption during navigation.
 - D) To detect dynamic objects and reclassify the environment.
- (b) (4 points) What is the main idea behind PRM?
 - A) Grow a tree from the start to the goal position
 - B) Randomly sample free space and connect samples to build a graph
 - C) Search the entire configuration space exhaustively
 - D) Maximize the robot's velocity during planning
- (c) (4 points) In PRM, which of the following is true about the roadmap?
 - A) It only connects samples in collision
 - B) It connects nearby collision-free samples
 - C) It only connects samples after reaching the goal
 - D) It requires a full map before planning starts
- (d) (4 points) What is the main advantage of RRT over PRM?
 - A) RRT builds a global roadmap in advance
 - B) RRT grows a tree quickly toward unexplored areas
 - C) RRT guarantees finding the shortest path
 - D) RRT only works in static environments
- (e) (4 points) In RRT, what happens when a random sample is generated?
 - A) It is ignored unless it is at the goal
 - B) It is connected to the nearest node in the tree if collision-free
 - C) It replaces the previous nodes in the tree
 - D) It deletes earlier branches of the tree
- (f) (4 points) Which of the following statements about PRM and RRT is correct?
 - A) Both PRM and RRT guarantee an optimal path.
 - B) PRM builds a graph first; RRT grows a tree during search.
 - C) RRT is slower than PRM in open environments.
 - D) PRM is only used for real-time dynamic environments.

5. Markov Decision Processes.

- (a) (2 points) In an MDP, what does the "Markov property" mean?
- A) The system always returns to its initial state.
 - B) The future depends only on the current state and action.
 - C) All rewards must be positive.
 - D) The transition probabilities never change.
- (b) (4 points) What does a "policy" represent in an MDP?
- A) A method for generating random states
 - B) A sequence of states
 - C) A mapping from states to actions
 - D) A method for calculating reward values
- (c) (4 points) What is the goal when solving an MDP?
- A) Find the fastest way to terminate the process.
 - B) Find a policy that maximizes cumulative expected reward.
 - C) Minimize the number of states visited.
 - D) Avoid high-reward states.
- (d) (4 points) In MDPs, what is "discounting" used for?
- A) Making immediate rewards more important than distant rewards.
 - B) Increasing the future rewards' value.
 - C) Ignoring all rewards beyond a fixed number of steps.
 - D) Penalizing the robot for collisions.
- (e) (4 points) Write the Bellman equation for the value of a state $V(s)$. What is the main idea behind the Bellman equation (1 sentence)?
- (f) (4 points) How does "policy iteration" differ from "value iteration"?